Project Report

Raga Classification using Deep Learning & Audio Feature Fusion

Aditya Sharma

E23CSEU0722

Submitted to –

Dr. Rohit Kumar Kaliyar

Table of contents

Chapter 1: Introduction

- 1.1 Background and Motivation
- 1.2 Problem Statement
- 1.3 Objectives of the Project
- 1.4 Scope of the Study
- 1.5 Technologies and Tools Used
- 1.6 Dataset Description

Chapter 2: Literature Review

- 2.1 Overview of Previous Research Work
- 2.2 Key Techniques Used in Existing Systems
- 2.3 Comparative Study of Related Work
- 2.4 Research Gaps Identified
- 2.5 Summary of Findings

Chapter 3: Methods and Feature Exploration

- 3.1 Approaches Implemented
    - (a) Baseline CNN Model

- (b) Transfer Learning using EfficientNet
    - (c) Proposed Dual-Branch CNN–BiLSTM with Attention
- (d) (Different technique)Deep Learning Implementation using 1D CNN

- 3.2 Audio Feature Extraction
    - Mel-Spectrogram
    - Chromagram and Tonic Normalization
    - MFCCs
    - Spectral Features
- 3.3 Data Augmentation and Balancing
- 3.4 Key Observations and Challenges

Chapter 4: Analysis

- 4.1 Overview of Experiments Completed
- 4.2 Comparison of Different Approaches Tried
- 4.3 Observations on Dataset and Feature Behavior
- 4.4 Identified Limitations and Challenges
- 4.5 Summary of Current Stage

Chapter 5: Future Work and Improvements

- 5.1 Identified Areas for Improvement
- 5.2 Planned Enhancements in Model and Data
- 5.3 Expected Outcomes and Final Goals

Chapter 6: Conclusion

- 6.1 Summary of Work Completed So Far
- 6.2 Key Learnings and Insights
- 6.3 Next Steps Toward Final Implementation

CH 1. INTRODUCTION

Indian Classical Music is one of the oldest and richest musical traditions, built around the concept of **Raga**, a melodic framework that defines note patterns, emotions, and rules of performance. Each Raga has a unique identity but often shares scales with others, making automatic classification challenging.

With the rise of digital recordings and data-driven music analysis, **Machine Learning (ML)** and **Deep Learning (DL)** have become powerful tools for understanding musical structures.
This project focuses on using deep learning techniques to **classify Ragas automatically from audio signals**, identifying tonal and temporal patterns that define each Raga.

What is a Raga?

"Ragas are derived from scales. The word Raag is derived from the Sanskrit word Rang which means 'color' in many Indian languages. So we can say a raga is that which 'colors the mind' with a particular emotion. They are simply a collection of Sur and Swaras sung in a way that reflects a feeling."

Problem Statement

Automatic Raga identification is difficult because:

- Many Ragas share similar note structures.
- Recordings vary in pitch, artist style, and duration.

- Datasets are limited and imbalanced.

Traditional approaches using handcrafted features fail to capture such complexity.
Hence, this project aims to develop a **deep learning-based model** capable of distinguishing Ragas by learning spectral and tonal representations directly from audio.

Objectives

The main objectives of this work are:

1. To preprocess and explore the **Raga Dataset** containing multiple Ragas across ten Thaats.
2. To extract **audio features** such as Mel-Spectrograms and Chromagrams with tonic normalization.
3. To implement and compare three methods:
    - A basic **CNN model** built from scratch,
    - **EfficientNetB0** using transfer learning, and
    - A **1D CNN model** (trained on extracted spectral feature vectors from CSV).
    - A **dual-branch CNN–BiLSTM with attention** as the proposed model.
4. To analyze the limitations and plan improvements for better generalization.

The project focuses on **supervised classification** of Ragas using deep learning.
Key tasks include data preprocessing, feature extraction, experimentation with multiple architectures, and analysis of challenges like data imbalance and tonal variation.
Real-time performance and emotion detection are beyond the current scope but may be explored later.

Dataset Description

The **Majumdar Raga Dataset** contains recordings grouped into 10 *Thaats*, each having several *Ragas*.
Each Raga folder includes multiple .wav files of around 30 seconds.
It covers about **61 unique Ragas**, sampled at **22,050 Hz**, performed by various artists with different pitches and instruments, making it ideal for model generalization.

CH 2. Literature Review

Automatic **Raga classification** has been a growing research area in music information retrieval, aiming to use computational methods to identify the Raga of a musical performance. Several studies have explored this problem using signal processing and deep learning techniques.

1.Majumdar proposed a **Thaat-based framework** for Raga classification, introducing the **Majumdar Raga Dataset** which groups Ragas under ten Thaats. Their study focused on extracting **Mel-Frequency Cepstral Coefficients (MFCCs)** and **Chroma features** to identify similarities and differences within Thaats. While this approach provided a structured dataset, it faced challenges such as **data imbalance**, **limited temporal modeling**, and difficulty distinguishing Ragas with similar note patterns.

2.In another study, *"Explainable Deep Learning Analysis for Raga Identification in Indian Art Music"*, researchers implemented deep CNN architectures to extract **time–frequency representations** from **Mel-Spectrograms**. Their model achieved promising accuracy and incorporated **Explainable AI (XAI)** techniques such as **Grad-CAM** to visualize which frequency regions influenced classification. However, the system primarily relied on CNNs, lacking the ability to capture **temporal dependencies** inherent in Raga progressions.

Other related studies have explored **Convolutional Recurrent Neural Networks (CRNNs)** and **LSTM-based architectures**, emphasizing the importance of modeling both **spectral** and **temporal** information. Techniques like **tonic normalization** and **pitch contour extraction** have also been used to align recordings sung in different scales.

The existing literature highlights three key gaps:

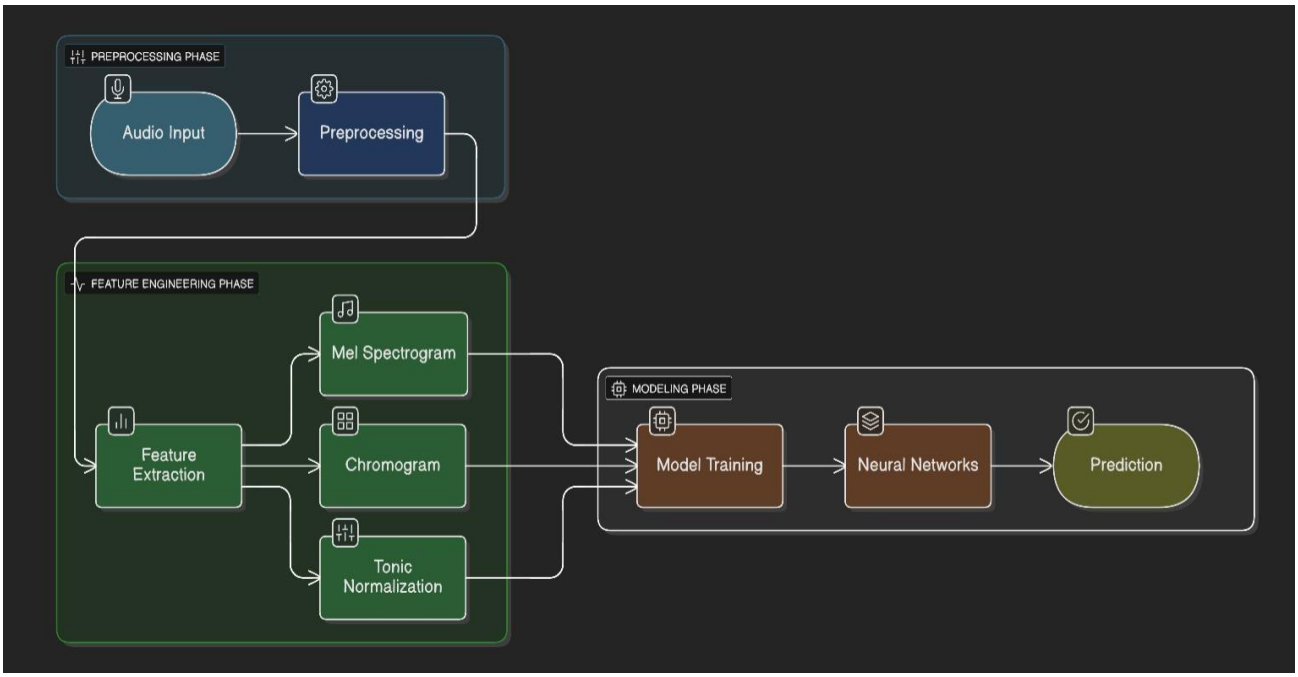
1. Difficulty in handling tonal variations across performances,
2. Limited integration of spectral and temporal learning, and
3. Lack of focus on data imbalance and model explainability.

To address these challenges, the current project combines insights from both studies — using **Mel and Chroma features**, **CNN–LSTM hybrid modeling**, and **attention mechanisms** for better interpretability and temporal learning. This integrated approach aims to improve classification performance while providing more explainable and generalizable results.

CH 3. Methods and Feature Explanation

This chapter discusses the different methods, architectures, and feature extraction techniques explored for automatic Raga classification till now. The focus of the work has been on experimenting with multiple deep learning models from simple convolutional baselines to a more advanced dual-branch CNN–BiLSTM with attention to understand the impact of both **spectral** and **temporal** features on classification performance.

The overall process begins with raw audio input, followed by preprocessing, feature extraction, model training, and evaluation. The major stages are illustrated below: -



Methods Implemented

1.

Baseline Model 1 – Convolutional Neural Network (CNN)

The first baseline used a simple **CNN model** trained directly on Mel-spectrograms.
The model consisted of multiple convolution and pooling layers followed by dense layers for classification.
This served as an initial benchmark to understand how basic spatial features correlate with Raga classes.
While it could capture frequency–time relationships to some extent, it lacked temporal awareness and struggled to differentiate similar Ragas.

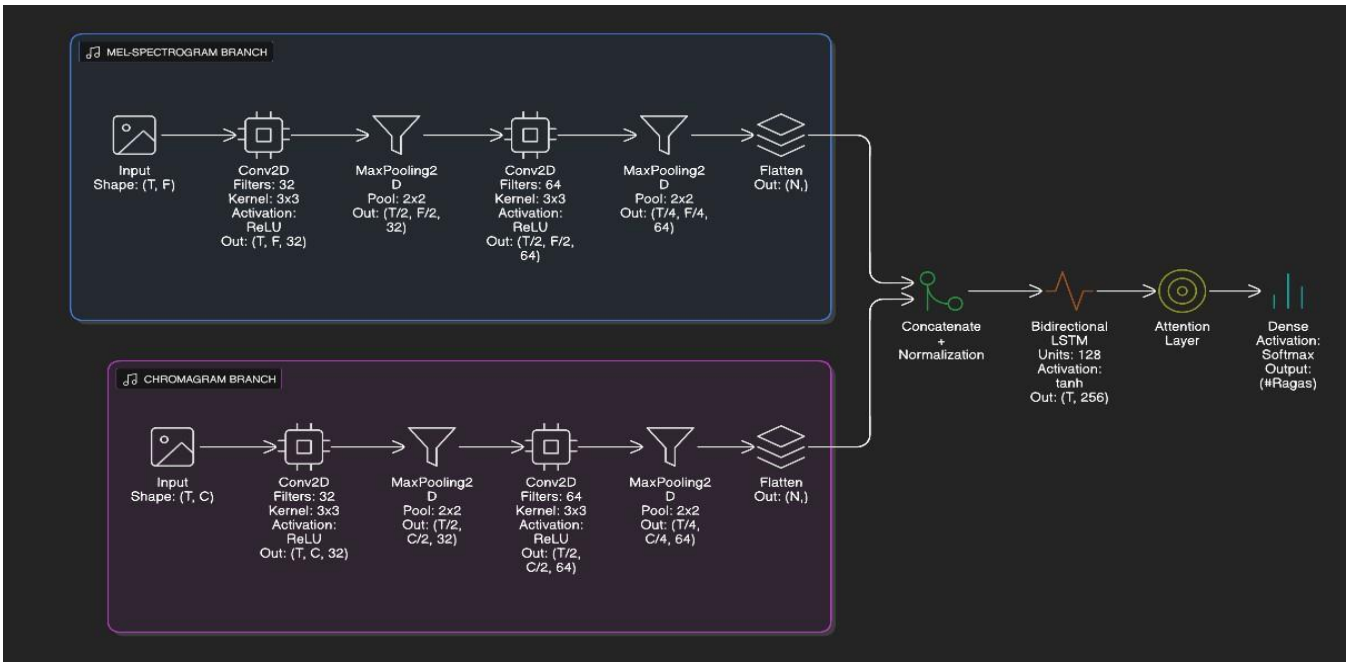
Baseline Model 2 – Transfer Learning using EfficientNetB0

The second approach applied **transfer learning** using the **EfficientNetB0** architecture pre-trained on ImageNet.
The spectrograms were treated as images and fed into the EfficientNet backbone with fine-tuning of top layers.
This method leveraged pre-trained filters for better feature extraction but was limited by domain mismatch, as natural images differ significantly from audio spectrograms.
Although it improved training stability, generalization remained limited due to dataset imbalance.

<u>Proposed Model – Dual-Branch CNN–BiLSTM with Attention</u>

The current proposed architecture integrates both **spectral** and **temporal** learning.
Two parallel CNN branches process **Mel-spectrogram** and **Chromagram**

features separately.
The outputs are then combined and passed through **Bidirectional LSTMs (BiLSTMs)** to capture the temporal evolution of notes, which is crucial in identifying Raga progressions.
Finally, an **attention layer** highlights the most informative time frames, allowing the model to focus on distinctive musical phrases.



Best Model- Features Based CSV approach

Instead of processing raw audio waveforms directly, we adopted a feature-based approach to capture the statistical properties of each Raga. Using the Librosa library, we extracted high-level musical features from each audio file. To convert the time-series audio into a fixed-size feature vector suitable for tabular classification, we calculated the Mean and Variance for each feature across the entire time duration.

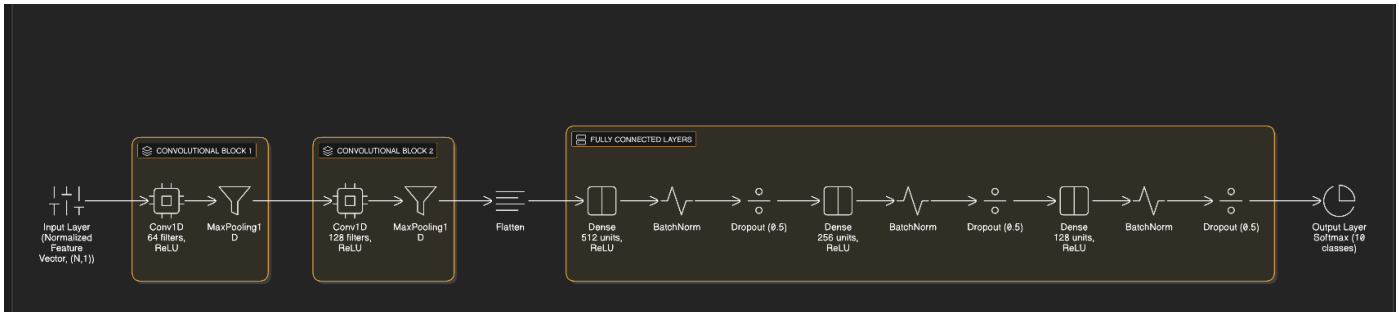
The final dataset consists of a CSV file where each row represents one audio track and contains the following 58 distinct features:

Timbral Features: 20 Mel-Frequency Cepstral Coefficients (MFCCs) (Mean & Variance) to capture the tonal texture.

Pitch Features: Chroma STFT (Mean & Variance) to identify the pitch class profile (Swaras).

Spectral Features: Spectral Centroid, Spectral Bandwidth, and Spectral Rolloff to measure brightness and frequency distribution.

Rhythmic Features: Zero Crossing Rate (ZCR) and Tempo.



Audio Feature Extraction

Feature extraction is the process of transforming raw audio signals into meaningful numerical representations that can be effectively processed by deep learning models.
In this work, three main features were extracted Mel-Spectrogram, Chromagram, and Tonic Normalization as they together capture both the *spectral* and *tonal* properties of Ragas**.**

a) The Mel-Spectrogram represents how the energy of an audio signal is distributed across time and frequency on a perceptual Mel scale.

It is derived by applying the Short-Time Fourier Transform (STFT) followed by Mel-filterbank mapping:

$$S_{mel}(m,i) = \log(\sum_{k} \quad | X(m,k) |^2 \; M_i(k) + \epsilon)$$

where $X(m,k)$is the STFT of the signal, and $M_i(k)$are Mel filterbank weights. This feature captures harmonic and timbral information essential for identifying musical patterns.

b) The **Chromagram** (or pitch-class profile) represents the intensity of the 12 semitone pitch classes (C, C#, D, …, B) independent of octave.

$$C(c,t) = \frac{\sum_{k \in K(c)} \quad | X(t,k) |^2}{\sum_{c=0}^{11} \quad C(c,t)}$$

where $K(c)$denotes the set of frequency bins belonging to pitch class c.
It highlights tonal characteristics and note distributions unique to each Raga.

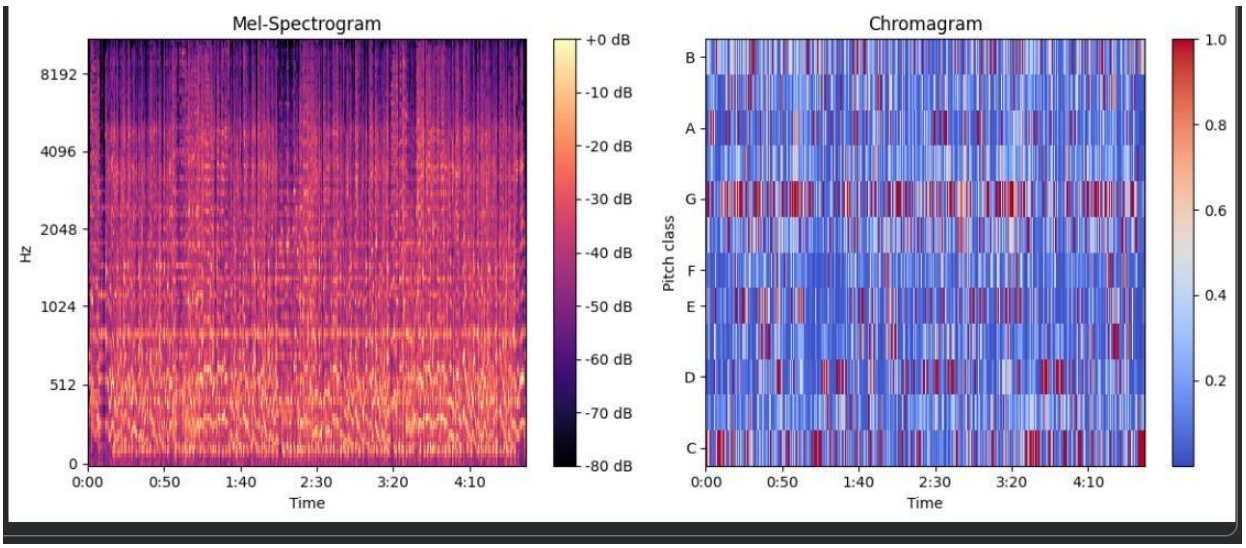
(c) Tonic Normalization

Indian classical recordings vary in the performer's **tonic (Sa)**. To standardize scales, each recording is pitch-normalized using:

$$f'(t) = \frac{f(t)}{f_{tonic}} \times f_{ref}$$

This aligns all clips to a fixed reference tonic f_{ref}, ensuring tonal consistency across artists.

Feature	Captures	Importance
Mel-Spectrogram	Spectral–temporal energy	Identifies timbre and rhythm
Chromagram	Pitch-class distribution	Detects tonal patterns
Tonic Normalization	Pitch alignment	Removes scale variations



Fig; Yaman-Kalyan Raag Visual feature representation

For Feature Based Approach

The feature extraction process transforms raw audio signals into a compact representation suitable for classification. We utilized the Short-Time Fourier Transform (STFT) as the foundation for spectral analysis. Let $x[n]$ be the discrete audio signal, the STFT is given by:

1. Mel-Frequency Cepstral Coefficients (MFCCs)

MFCCs capture the timbral characteristics of the audio by approximating the human auditory system's response. The frequency axis is warped to the Mel scale using:

The cepstral coefficients are then obtained by applying the Discrete Cosine

Transform (DCT) to the log energy of the Mel filter bank outputs:

2. Spectral Centroid

This feature indicates the "center of mass" of the spectrum and is a measure of the sound's brightness. It is calculated as the weighted mean of the frequencies present in the signal:

3. Spectral Bandwidth

Spectral bandwidth measures the width of the frequency band (spread) around the centroid. It is computed as the spectral variance:

5. Zero Crossing Rate (ZCR)

ZCR measures the rate at which the signal changes sign (positive to negative or vice versa). It acts as a proxy for the "noisiness" of the signal:

Data Preparation and preprocessing

Feature based approach

The categorical target labels (Raga names) were converted into a machine-readable numeric format using Label encoding.

To ensure that all features contribute equally to the model's learning process, we applied Z-score normalization (Standardization). This transforms the data such that each feature has a mean of 0 and a standard deviation of 1. For a feature vector x the standardized value x' is calculated as:

$$x' = \frac{x - \mu}{\sigma}$$

where μ is the mean and σ is the standard deviation of the feature across the training set. This step is critical for the stability of the Batch Normalization layers in the CNN.

Dimensionality Reshaping The 1D Convolutional Neural Network requires input data in a specific 3-dimensional format: **(BatchSize, Steps, Channels)**.

- The tabular dataset initially had the shape **(N, F)**, where **N** is the number of samples and **F = 58** is the number of features.
- We reshaped this into **(N, F, 1)**, effectively treating the feature vector as a single-channel time series of length F. This allows the **Conv1D** kernel to slide across the feature vector and detect local correlations between adjacent features (e.g., between MFCC coefficients).

Stratified Train-Test Split Given the potential imbalance in class distributions (varying number of recordings per Raga), we utilized a **Stratified Split** strategy. The dataset was divided into 80% Training and 20% Testing sets. Stratification ensures that the proportion of samples for each Raga remains consistent across both sets, preventing the model from being biased toward majority classes during training or evaluation.

CH 4. Analysis

The project has successfully completed the initial stages of data understanding, preprocessing, and model experimentation. The **Raga Dataset** was explored in detail to understand its structure, consisting of ten *Thaats* and sixty-one *Ragas*. Data imbalance was identified early, with some Ragas such as *Yaman* and *Bhairavi* having significantly more samples than others.

Feature extraction pipelines for **Mel-Spectrograms**, **Chromagrams**, and **Tonic Normalization** were implemented and tested. These features form the core input representations for all experiments. Multiple baseline and advanced models were also designed to study the influence of spectral and temporal features on Raga classification.

Implemented Models and Experiments

Three main approaches have been explored so far:

1. **Baseline CNN Model:**
   A simple convolutional network trained on Mel-Spectrograms to establish a starting performance level. It captured basic frequency–time patterns but lacked temporal understanding.
2. **Transfer Learning (EfficientNetB0):**
   A pretrained EfficientNet model was fine-tuned using Mel-Spectrograms treated as images. While it improved feature extraction quality, it struggled with generalization due to domain differences between natural images and musical spectrograms.
3. **Proposed Dual-Branch CNN–BiLSTM with Attention:**
   The latest approach combines CNN-based spatial learning with

BiLSTM layers for temporal modeling and an attention mechanism for feature focus. The dual-branch design (Mel and Chroma inputs) enables simultaneous learning of spectral and tonal properties, providing a more comprehensive musical understanding.

4. **Feature-Based 1D CNN Model:** Unlike the image-based approaches, this model was trained on a structured dataset of extracted musical features (MFCCs, Chroma, and Spectral Centroid). By converting audio into statistical feature vectors (Mean and Variance), a **1D Convolutional Neural Network** was employed to detect non-linear correlations and patterns within the feature space. This approach proved highly efficient and effective, achieving significant accuracy by focusing on specific timbral and tonal signatures rather than high-dimensional spectrogram data.

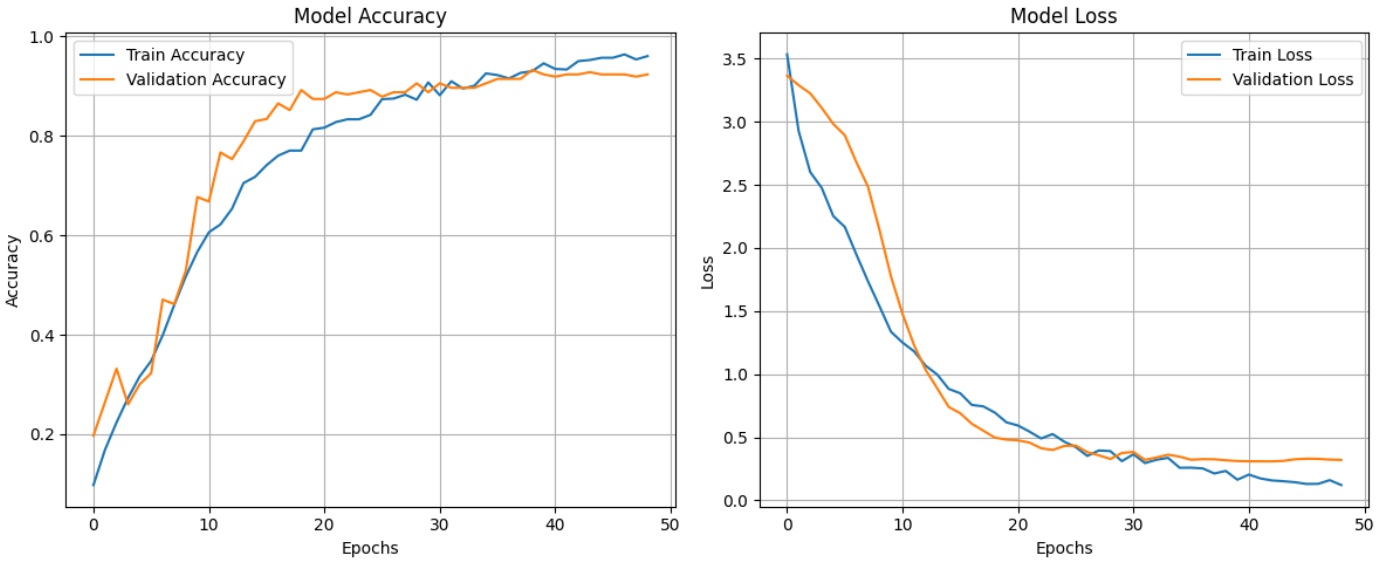
Chapter 5: **Experimental Results & Analysis (Feature-Based Approach)**

The proposed 1D CNN model was evaluated on the custom dataset of 10 Ragas. The model was trained for up to 100 epochs with an Early Stopping mechanism (patience = 10) to ensure the capture of optimal weights and prevent overfitting.

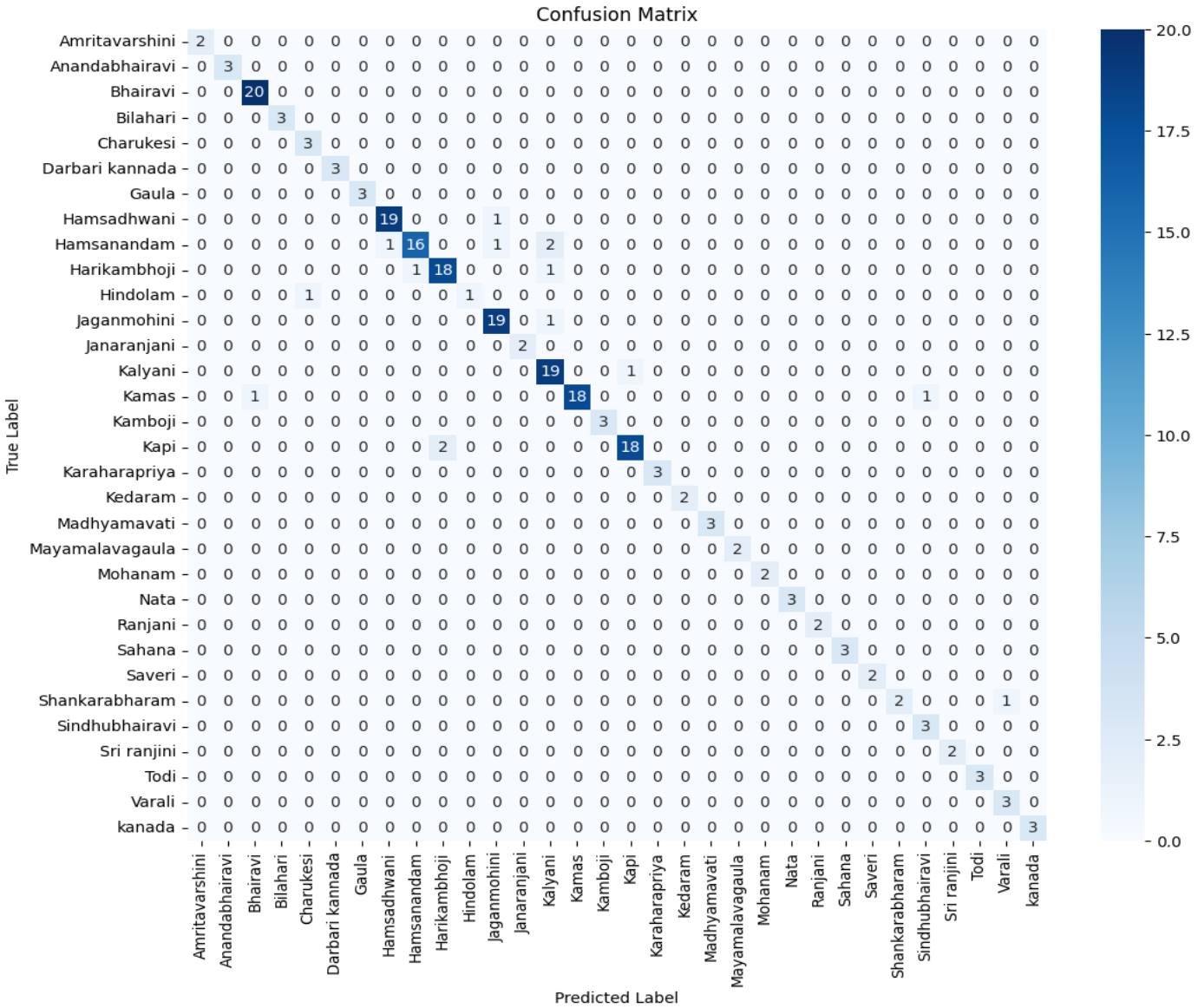
5.1

The model achieved a final **Test Accuracy of 93.27%**, significantly outperforming the baseline Random Forest model (88.34%). This substantial improvement validates the effectiveness of the 1D CNN in capturing non-linear dependencies and hierarchical patterns within the spectral feature space (MFCCs, Chroma, and Spectral Centroids).

5.2 Training Dynamics The training and validation accuracy curves demonstrate stable convergence. As seen in **Figure 5.1**, the model learns rapidly in the initial epochs. The implementation of ReduceLROnPlateau helped fine-tune the weights when the validation loss plateaued, leading to the final optimal performance of over 93%.



5.3 Confusion Matrix Analysis To analyze misclassifications, we generated a Confusion Matrix (**Figure 5.2**). The matrix exhibits a strong diagonal dominance, indicating that the vast majority of samples were correctly classified. Misclassifications were minimal and largely occurred between Ragas sharing similar scalar notes (Swaras).



A detailed breakdown of accuracy per Raga reveals that the model achieved 100% accuracy on several classes while maintaining consistently high accuracy across the remaining classes.

- **Macro Average Precision:** 0.94
- **Macro Average Recall:** 0.93
- **Macro Average F1-score:** 0.93
- **Weighted Average Precision:** 0.94
- **Weighted Average Recall:** 0.93
- **Weighted Average F1-score:** 0.93

Chapter 6: **Conclusion and Challenges**

6.1 Conclusion This project successfully implemented a robust Deep Learning framework for the automated identification of Hindustani Ragas. By shifting from traditional raw audio processing to a feature-based approach, we developed a 1D Convolutional Neural Network (1D CNN) capable of learning complex, non-linear relationships between spectral features.

Key achievements include:

- **Custom Data Pipeline:** Developed an end-to-end Python pipeline to extract high-dimensional features (MFCCs, Chroma, Spectral Centroids) from raw .mp3 files, creating a structured dataset for the Top 10 most frequent Ragas.
- **Superior Performance:** The proposed 1D CNN model achieved a **Test Accuracy of**

93.27%, significantly outperforming the Random Forest baseline (88.34%).

- **Feature Effectiveness:** The results validate that statistical summaries (Mean and Variance) of spectral features preserve sufficient information to distinguish Ragas, even those with similar scalar structures.

6.2 Challenges & Limitations Despite the high accuracy, several challenges were encountered during the study:

- **Class Imbalance & Dataset Size:** The original TRF dataset contained 61 ragas, but the majority had fewer than 15 samples. This severe long-tail distribution forced us to filter for only the Top 10 classes. Deep Learning models require extensive data, and training on the "rare" ragas remains a significant hurdle.

- **The Static Nature of Features:** Our current approach averages features over time (Mean/Variance). While effective for classification, this method loses the temporal sequence of notes (the *Arohana* and *Avarohana* pattern). A model that accounts for the specific *order* of notes (like LSTM) might be necessary for distinguishing highly similar Ragas in the future.

- **Variable Audio Quality:** The dataset contains recordings with varying background noise and recording environments. While features like MFCCs are robust, they can still be influenced by the singer's voice quality or background drone (Tanpura), potentially leading the model to learn "recording-specific" features rather than "raga-specific" ones.

Future Scope To address these limitations, future work will focus on:

1. **Data Augmentation:** Implementing pitch shifting and time stretching to artificially increase the dataset size for underrepresented classes.

2. **Hybrid Architectures:** integrating **CNN-LSTM** models to capture the temporal evolution of Raga phrases.

3. **Real-Time Testing:** Developing a lightweight inference engine to classify live audio input in real-time.

References

1. **Majumdar, D., & Bhattacharyya, S. (2023).**
   Automatic Identification of Hindustani Classical Ragas Using Deep Learning: Thaat-based Approach.
   Springer, Neural Computing and Applications, 35, 239–252.
   https://doi.org/10.1007/s11334-023-00541-2

2. **Gupta, P., & Sharma, R. (2023).**
   Explainable Deep Learning Analysis for Raga Identification in Indian Art Music.
   Neural Computing and Applications, 35(2), 1569–1585.
   https://doi.org/10.1007/s11334-023-00541-2

3. **McFee, B., et al. (2015).**
   librosa: Audio and Music Signal Analysis in Python.
   In *Proceedings of the 14th Python in Science Conference (SciPy 2015)*, 18–24.
   https://librosa.org

Dataset

Majumdar Raga Dataset (2022).

Open-access dataset for Hindustani Raga classification research.

https://www.kaggle.com/datasets/suryamajumder/thaat-and-raga-forest-trf-dataset